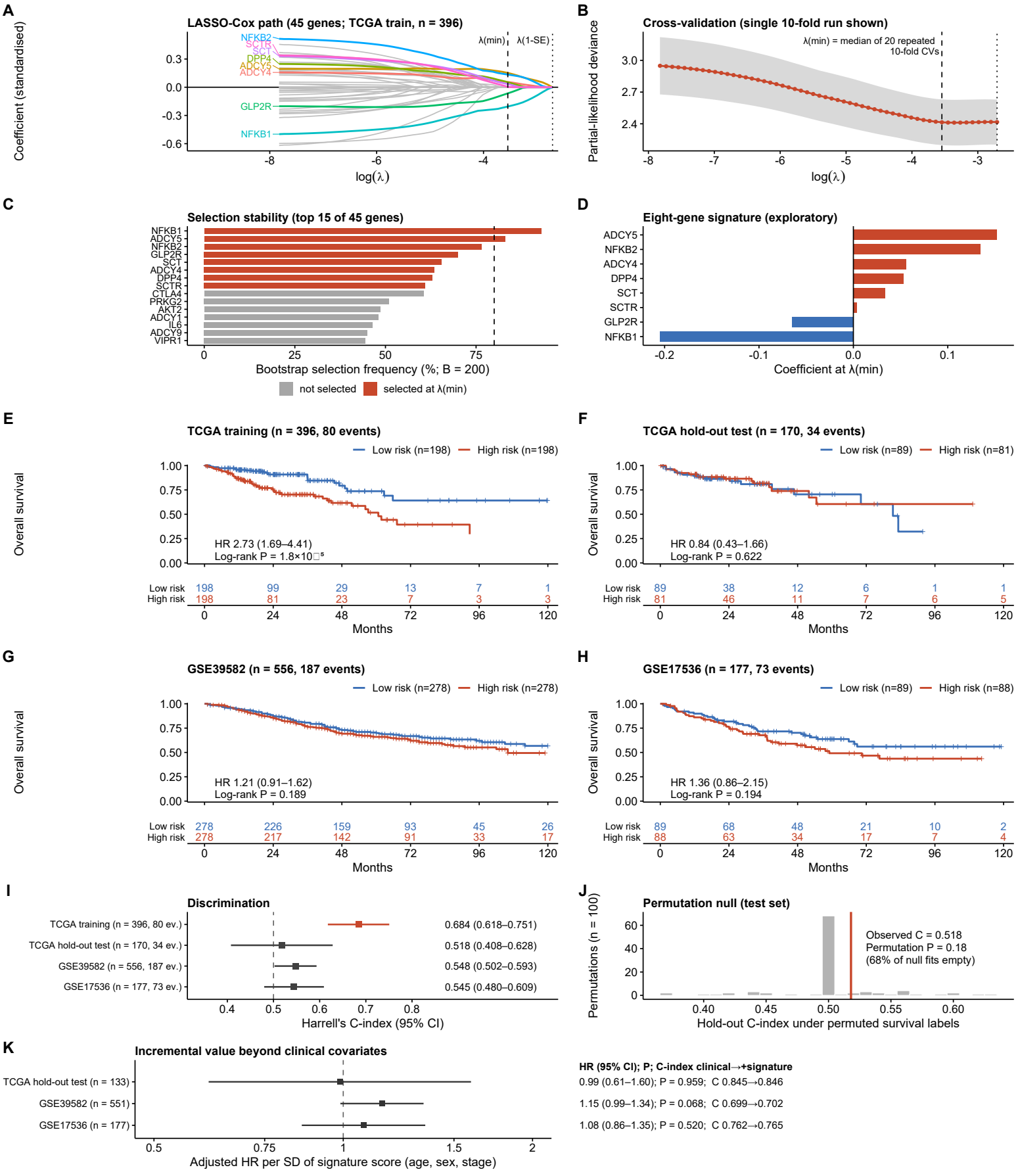

A
LASSO-Cox path (45 genes; TCGA train, n = 396)
Coefficient (standardised)
log(λ)
λ(min)
λ(1-SE)
NFKB2
SCTR
SCT
DPP4
ADCY5
ADCY4
GLP2R
NFKB1
B
Cross-validation (single 10-fold run shown)
Partial-likelihood deviance
log(λ)
λ(min) = median of 20 repeated 10-fold CVs
C
Selection stability (top 15 of 45 genes)
Bootstrap selection frequency (%; B = 200)
not selected
selected at λ(min)
D
Eight-gene signature (exploratory)
Coefficient at λ(min)
E
TCGA training (n = 396, 80 events)
Low risk (n=198)
High risk (n=198)
Overall survival
HR 2.73 (1.69–4.41)
Log-rank P = 1.8×10⁻⁵
Months
F
TCGA hold-out test (n = 170, 34 events)
Low risk (n=89)
High risk (n=81)
HR 0.84 (0.43–1.66)
Log-rank P = 0.622
G
GSE39582 (n = 556, 187 events)
Low risk (n=278)
High risk (n=278)
HR 1.21 (0.91–1.62)
Log-rank P = 0.189
H
GSE17536 (n = 177, 73 events)
Low risk (n=89)
High risk (n=88)
HR 1.36 (0.86–2.15)
Log-rank P = 0.194
I
Discrimination
TCGA training (n = 396, 80 ev.)
TCGA hold-out test (n = 170, 34 ev.)
GSE39582 (n = 556, 187 ev.)
GSE17536 (n = 177, 73 ev.)
0.684 (0.618–0.751)
0.518 (0.408–0.628)
0.548 (0.502–0.593)
0.545 (0.480–0.609)
Harrell's C-index (95% CI)
J
Permutation null (test set)
Permutations (n = 100)
Observed C = 0.518
Permutation P = 0.18
(68% of null fits empty)
Hold-out C-index under permuted survival labels
K
Incremental value beyond clinical covariates
TCGA hold-out test (n = 133)
GSE39582 (n = 551)
GSE17536 (n = 177)
Adjusted HR per SD of signature score (age, sex, stage)
HR (95% CI); P; C-index clinical→+signature
0.99 (0.61–1.60); P = 0.959; C 0.845→0.846
1.15 (0.99–1.34); P = 0.068; C 0.699→0.702
1.08 (0.86–1.35); P = 0.520; C 0.762→0.765